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Law Enforcement’s Use of Facial Recognition Technology

Statement for the Record

Good afternoon Chairman Chaffetz, Ranking Member Cummings, and members of the committee. Thank you for the opportunity to appear before you today to discuss the Federal Bureau of Investigation's (FBI) use of face recognition (FR) technology.

FBI Programs Perform Face Recognition

The following FBI programs use FR technology for law enforcement purposes.¹ They are: (1) the FBI's Next Generation Identification (NGI) System located at the FBI's Criminal Justice Information Services (CJIS) Division, and (2) the Facial Analysis, Comparison, and Evaluation (FACE) Services Unit also located at the FBI CJIS Division.

1. **NGI maintains a mugshot repository that is known as the Interstate Photo System (IPS).** All mugshots are associated with tenprint fingerprints and a criminal history record. The NGI-IPS allows automated FR searches by authorized local, state, tribal, and federal law enforcement agencies. The law enforcement agency submits a “probe” photo that is obtained pursuant to an authorized law enforcement investigation, to be searched against the mugshot repository. The NGI-IPS returns a gallery of “candidate” photos of 2-50 individuals (default is 20). The law enforcement agencies then must manually review the candidate photos and perform further investigation to determine if any of the candidate photos are the same person as the probe photo.

The NGI-IPS technology is only used as an investigative lead, and not as a means of positive identification. The NGI-IPS *Policy Implementation Guide* has been made available to authorized law enforcement users who receive candidate photos from the Next Generation Identification-Interstate Photo System. The policy advises that the photos are not being provided as positive identification and cannot serve as the sole basis for law enforcement action. In addition, the FBI has promulgated policies and procedures that place legal, policy, and security requirements on the law enforcement users of the NGI-IPS, including a prohibition against submitting probe photos that were obtained in violation of the First or Fourth Amendments. It is important to note that the FBI does not retain the probe photos; the probes are searched and deleted. Therefore, the NGI-IPS remains a repository solely of mugshots that are submitted voluntarily with fingerprints pursuant to arrest.

2. **The FACE Services Unit** provides investigative lead support to the FBI field offices, operational divisions, and legal attachés by comparing the face images of persons associated with open assessments² and active investigations³ against face images available in state and federal FR systems. In limited instances, the FACE Services Unit provides FR support for closed FBI cases (e.g., missing and wanted persons) and may offer recognition support to federal partners. The FACE Services Unit only accepts “probe” photos that have been collected pursuant to applicable legal authorities as part of an authorized FBI investigation. Upon receipt of the photo(s), the FACE Services Unit searches them using FR software against databases authorized for use by the FBI, which results in a photo gallery of potential candidates. The FACE Services Unit performs manual comparisons of candidate photos against the probe photo(s) to determine a candidate's value as an investigative lead. This service does NOT provide positive identification, but rather, an investigative lead and analysis to support that lead.

In performing the search(es), the FACE Services Unit operates under the authority of the United States Code (U.S.C) Sections 533 and 534; Title 28, Code of Federal Regulations Section 0.85; Title 42, U.S.C. Section 3771; and Title 18, U.S.C. Chapter 123. The FACE Services Unit performs FR searches of FBI databases (e.g., FBI's NGI), other federal databases (e.g., Department of State's Visa Photo File, Department of Defense's Automated Biometric Identification System, Department of State's Passport Photo File), and state photo repositories (e.g., select state Departments of Motor Vehicles). Memoranda of understanding and agreements have been established with all partners.

Privacy Impact Assessments (PIAs) for the FACE Services Unit and the NGI-IPS have been prepared by the FBI, approved by the DOJ, and posted at <https://www.fbi.gov/services/records-management/foipa/privacy-impact-assessments>. These PIAs provide to the public an accurate and complete explanation of how specific FBI components are using face recognition technology in support of the FBI's mission to defend against terrorism and enforce criminal laws, while protecting civil liberties. The PIAs also reflect many of the privacy and civil liberties choices made during the implementation of these programs.

Automated Face Recognition⁴

In addition to understanding how these FBI programs operate, it is also important to have an understanding of how automated FR works. The following is a brief description of automated FR: The automated FR software uses pattern matching approaches developed within the field of computer vision. Such approaches do not rely upon intrinsic models of what a face is, how it should appear, or what it may represent. In other words, the potential matching is not based on biological or anatomical models of what a face—or the features which make up a face—look like. Instead, the algorithm performance is entirely dependent upon the patterns which the algorithm developer finds to be most useful for matching. The patterns used in automated FR algorithms do not correlate to obvious anatomical features such as the eyes, nose or mouth in a one-to-one manner, although they are affected by these features. Put another way, the algorithms “see” faces in a way that differs from how humans see faces.

Accuracy

The FBI conducted a trade study of FR products, leveraging the NGI Integrator Lockheed Martin, which led to the determination of MorphoTrust as the best cost solution in fall 2010. The FBI has tested and verified that the NGI FR solution returns the correct candidate a minimum of 85 percent of the time within the top 50 candidates.

The FBI manages the CJIS Division Advisory Policy Board (APB) Process, which holds meetings twice a year. The APB is comprised of members of local, state, tribal, and federal criminal justice agencies that contribute to and use CJIS systems and information. It is responsible for reviewing policy issues and appropriate technical and operational issues related to FBI CJIS programs (such as the NGI) administered by the FBI's CJIS Division, and thereafter, making appropriate recommendations. Through the APB Process, users can provide feedback and suggestions or bring issues to the attention of the FBI's CJIS Division. To date, no users have submitted concerns to the FBI regarding the accuracy of face searches conducted on the NGI-IPS.

Audits

The FBI performs audits as they serve an important role in identifying and mitigating risks associated with users of information systems not meeting policy requirements. In a recent audit of the FBI's use of FR by the Government Accountability Office (GAO), the FBI advised that the NGI-IPS operated in a limited capacity as a pilot program from December 2011 through April 2015. While the early stages of planning for formal NGI-IPS audits began during the system's pilot phase and prior to GAO's review, the formal draft audit plan was completed on schedule in summer 2015 and approved by the CJIS APB in June 2016.